

STA630

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BY



EXPLORER

GROUP LINK;

<https://chat.whatsapp.com/FfCBn1AZMaB4NENQmgkJRD>

Q#1 Identify experimental design.

****Answer**:** Experimental design involves manipulating one or more independent variables to observe their effect on a dependent variable within a controlled environment. Key components include the treatment (independent

variable), outcome (dependent variable), control group, and random assignment of subjects to minimize biases.

Q#2 Steps for fieldwork

****Answer**:** The steps for fieldwork include: 1) Preparation, which involves planning the research design and logistics; 2) Data Collection, gathering data through observations, surveys, or interviews in the field; 3) Data Analysis, analyzing the collected data to draw conclusions; and 4) Reporting, presenting the findings in a structured format.

Q#3 What do you understand by the term exclusive and exhaustive?

****Answer**:** Exclusive means each data point fits into one and only one category, ensuring no overlap. Exhaustive means all possible data points are covered by the categories, ensuring completeness. Together, they ensure data categorization is clear and comprehensive.

Q#4 Quasi-experiment design

****Answer**:** Quasi-experiment design lacks random assignment of subjects to experimental and control groups. It uses existing groups and conditions, making it more practical in real-world settings where randomization is not feasible. This design helps assess causal relationships despite potential confounding variables.

Q#5 Loaded question and burdensome question difference

****Answer**:** A loaded question contains a controversial or unjustified assumption, potentially leading respondents toward a particular answer (e.g., "Have you stopped cheating?"). A burdensome question demands excessive recall

or detailed information that may be difficult for respondents to provide, making it challenging to answer accurately.

Q#6 Difference between structured and unstructured interview

****Answer**:** A structured interview uses a predefined set of questions with fixed answers, ensuring consistency and comparability of responses. An unstructured interview is more flexible, allowing the interviewer to explore topics in depth based on the respondent's answers, providing richer qualitative data.

Q#7 Explain Contrived non-contrived

****Answer**:** Contrived settings are artificial environments created by researchers to control variables and manipulate conditions. Non-contrived settings are natural environments where phenomena occur naturally without manipulation, allowing for observation of genuine behaviors and interactions.

Q#8 Explain Fieldwork

****Answer**:** Fieldwork is the process of collecting data outside of a laboratory, library, or workplace setting. It involves direct interaction with subjects or phenomena in their natural environment, using methods like observations, interviews, and surveys to gather authentic data.

Q#9 Explain Scale and indexes

****Answer**:** A scale measures the intensity, direction, or level of a variable, such as a Likert scale that gauges agreement levels. An index is a composite measure that summarizes and ranks specific observations or indicators, combining multiple variables into a single score.

Q#10 Explain Pre-testing and Post-testing

****Answer**:** Pre-testing involves conducting a trial run before the main study to refine data collection instruments or procedures. Post-testing is conducted after the intervention to assess its effects, comparing pre-intervention and post-intervention results to determine impact.

Q#11 Experimental

****Answer**:** An experimental study involves manipulating one or more independent variables to observe their effect on a dependent variable, typically with random assignment of subjects to control and experimental groups. This design helps establish causality by controlling extraneous variables.

Q#12 Probability and non-probability

****Answer**:** Probability sampling ensures every member of the population has a known and equal chance of being selected, enhancing representativeness. Non-probability sampling does not ensure equal chances for all members, often used when probability sampling is impractical, and may introduce bias.

Q#13 Pretest-posttest

****Answer**:** A pretest-posttest design measures the dependent variable before and after the treatment or intervention. This approach helps assess the effect of the intervention by comparing changes in the dependent variable from pre-intervention to post-intervention, indicating the treatment's impact.

Q#14 Method of conducting research

****Answer**:** Conducting research involves several steps: defining the problem, reviewing literature, forming hypotheses, designing the study, collecting data, analyzing data, and reporting results. These steps ensure a systematic approach to investigating and understanding the research question.

Q#15 Data transformation and its importance

****Answer**:** Data transformation involves converting data into a suitable format for analysis, such as normalizing or coding data. It is important for ensuring consistency, compatibility, and accuracy in data analysis, making it easier to interpret and draw valid conclusions from the data.

Q#16 Field experiment and lab experiment

****Answer**:** A field experiment is conducted in a natural setting, where variables and conditions are less controlled, providing real-world applicability. A lab experiment is conducted in a controlled, artificial setting, allowing precise manipulation of variables but potentially lacking external validity.

Q#17 Ethnography and ethnomethodology data collection

****Answer**:** Ethnography involves immersive observation and interaction within a community or culture to understand their practices and behaviors. Ethnomethodology studies the methods people use to understand and produce social order in everyday life, often through detailed observation and analysis.

Q#18 Researcher used a table for data presentation. Write any 3 parts of the table.

****Answer**:** The three main parts of a table for data presentation are: 1) Title, which describes the content and purpose of the table; 2) Headings, which define the columns and rows; and 3) Body, which contains the actual data and information presented in the table.

Q#19 If a researcher uses a survey interview method then write 3 ways to collect data.

****Answer**:** Three ways to collect data using a survey interview method are: 1) Face-to-face interviews, allowing for in-depth responses and clarification; 2) Telephone interviews, enabling data collection from remote participants; and 3) Online surveys, providing convenience and broader reach.

Q#20 Difference between pre-test and post-test

****Answer**:** A pre-test measures baseline data before an intervention, establishing a starting point for comparison. A post-test measures data after the intervention to assess its effects, comparing pre-test and post-test results to determine changes and the impact of the intervention.

Q#21 Difference between editing and data coding

****Answer**:** Editing involves reviewing and correcting collected data to ensure accuracy and completeness. Data coding assigns numerical or categorical codes to responses, facilitating efficient data entry and analysis by converting qualitative data into a quantitative format.

Q#22 Ahmad is doing research want to see the effect of temperature on productivity or performance. He increases the temperature after every 30 minutes and records the unit of production.

****Answer**:** Ahmad's research involves an experimental design. The independent variable is temperature, which he manipulates by increasing it every 30 minutes. The dependent variable is productivity or performance, measured by recording the units of production.

Q#23 Determine the treatment and dependent variables with reason

****Answer**:** The treatment variable is temperature, as it is the factor being manipulated in the experiment. The dependent variable is productivity or performance, as it is the outcome being measured to assess the effect of the temperature changes.

Q#24 Give suggestions for recording observation during an interview

****Answer**:** Suggestions for recording observations during an interview include:
1) Use a structured format to ensure consistency and completeness; 2) Take detailed notes, capturing key points and non-verbal cues; and 3) Utilize audio or video recording to allow for accurate review and analysis later.

Q#25 Significance of recording field notes in field research

****Answer**:** Recording field notes in field research is significant as they provide a detailed, real-time account of observations, which are crucial for data analysis and validation of findings. Field notes capture the context and nuances of the observed phenomena, enhancing the richness of the data.

Q#26 How Our exceptional services improved your overall work experience? Is it a leading or burdensome question with reason?

****Answer**:** The question "How Our exceptional services improved your overall work experience?" is a leading question because it assumes the services were exceptional and had a positive impact, potentially biasing the respondent's answer toward a favorable response.

Q#27 Scale vs. index

****Answer**:** A scale measures the intensity, direction, or level of a variable, such as a Likert scale that gauges agreement levels. An index is a composite measure that summarizes and ranks specific observations or indicators, combining multiple variables into a single score for analysis.

Q#28 Give the situation in which you prefer stratified random sampling over others and give any 3 reasons

****Answer**:** Stratified random sampling is preferred when: 1) The population has distinct subgroups; 2) Ensuring representation from each subgroup is important; and 3) Comparing specific subgroups is necessary. This method ensures accurate and reliable representation of all relevant groups.

Q#29 Ali wants to do an observation study in nature (field). Write down the steps he should follow.

****Answer**:** Steps Ali should follow for a field observation study include: 1) Define the objectives of the study; 2) Select the appropriate site for observation; 3) Plan the observation schedule to ensure systematic data collection; 4) Collect data systematically, recording observations accurately; and 5) Analyze and interpret the collected data.

Q#30 What is transformation and why is it important?

****Answer**:** Data transformation involves converting data into a suitable format for analysis, such as normalizing or coding data. It is important for ensuring consistency, compatibility, and accuracy in data analysis, making it easier to interpret and draw valid conclusions from the data.

Q#31 What is mean by mean, median, mode?

****Answer**:**

- ****Mean**:** The mean is the average of a data set, calculated by adding all the values together and dividing by the number of values.
- ****Median**:** The median is the middle value in a data set when the values are arranged in ascending or descending order. If there is an even number of values, it is the average of the two middle values.
- ****Mode**:** The mode is the value that appears most frequently in a data set. There can be more than one mode if multiple values have the same highest frequency.

Q#32 What happened in the absence of pilot testing?

****Answer**:** Without pilot testing, the research tools may contain flaws, leading to inaccurate or unreliable data. Issues like ambiguous questions, technical problems, or unforeseen difficulties may go undetected, affecting the validity and reliability of the study results.

Q#33 Similarity of Historical-comparative and fieldwork research

****Answer**:** Both historical-comparative and fieldwork research involve detailed, qualitative data collection and analysis. They aim to understand complex social phenomena in context and often require interpretative approaches to derive meaning from data.

Q#34 Five suggestions for accurate data recording

****Answer**:** Suggestions for accurate data recording include: 1) Use standardized forms or software; 2) Double-check entries for accuracy; 3) Record data immediately to avoid forgetting details; 4) Maintain consistent coding schemes; and 5) Regularly back up data to prevent loss.

Q#35 Do you agree sample accessibility through email has some disadvantages?

****Answer**:** Yes, sample accessibility through email can have disadvantages such as potential non-response bias, limited reach to individuals without internet access, and concerns about privacy and data security, which may affect the representativeness and reliability of the sample.

Q#36 Disadvantage of content analysis research

****Answer**:** Content analysis research can be time-consuming and labor-intensive, especially when analyzing large volumes of text. It may also be subjective, as coding decisions can be influenced by the researcher's biases, potentially affecting the study's validity and reliability.

Q#37 Survey research

****Answer**:** Survey research involves collecting data from a predefined group of respondents to gain information about their opinions, behaviors, or characteristics. It can be conducted through various methods such as questionnaires, interviews, or online surveys, offering versatility and scalability.

Q#38 Internal validity

****Answer**:** Internal validity refers to the extent to which a study accurately measures the causal relationship between variables, free from confounding factors. High internal validity means that observed effects are genuinely due to the independent variable, not other extraneous variables.

Q#39 Types of non-probability

****Answer**:** Types of non-probability sampling include: 1) Convenience sampling, selecting subjects who are easily accessible; 2) Judgmental (purposive) sampling, selecting subjects based on specific criteria; and 3) Snowball sampling, using existing subjects to recruit more participants.

Q#40 Limitations of historical-comparative research

****Answer**:** Limitations include potential bias in historical records, difficulty in establishing causality, and the challenge of generalizing findings from specific historical contexts to other settings. Interpretation can also be subjective, influenced by the researcher's perspective.

Q#41 Parameter and statistics

****Answer**:** A parameter is a numerical characteristic of a population, such as a population mean. A statistic is a numerical characteristic of a sample, used to estimate the corresponding population parameter. Parameters are fixed, while statistics vary with different samples.

Q#42 Maturation and testing

****Answer**:** Maturation refers to changes in subjects over time that can affect study outcomes, independent of the treatment. Testing effects occur when repeated testing influences participants' performance, potentially skewing the results by introducing practice effects or sensitization.

Q#43 Historical-comparative research

****Answer**:** Historical-comparative research examines past events and compares them across different contexts to understand social phenomena and trends. It involves analyzing historical documents, artifacts, and other sources to draw parallels and differences across time and place.

Q#44 Difficulty in establishing causality

****Answer**:** Establishing causality is difficult due to confounding variables that can influence the observed relationship between the independent and dependent variables. Experimental control, randomization, and longitudinal studies can help, but ethical and practical constraints often complicate this process.

Q#45 How does the internet capture real-time data? What do you think? Ethnography and ethnomethodology?

****Answer**:** The internet captures real-time data through automated tools like web scraping, APIs, and sensors. Ethnography involves immersive, in-depth study of cultures and communities. Ethnomethodology examines everyday social interactions to understand how people construct social reality.

Q#46 Pilot testing

****Answer**:** Pilot testing involves conducting a preliminary study to identify and resolve potential issues before the main research. It helps refine instruments, procedures, and logistics, ensuring the study is feasible, reliable, and valid. It also provides valuable feedback for improvement.

Q#47 Difference between simple sampling and systematic sampling

****Answer**:** Simple random sampling gives each population member an equal chance of selection, ensuring unbiased representation. Systematic sampling selects every n th member from a list, starting from a random point, offering ease of use while maintaining representativeness if the list order is random.

Q#48 Difference between control group in a research study and experimental group

****Answer**:** The control group does not receive the treatment or intervention, serving as a baseline for comparison. The experimental group receives the treatment or intervention. Comparing outcomes between these groups helps determine the treatment's effect, controlling for confounding variables.

Q#49 Discuss reasons for personal interview

****Answer**:** Personal interviews allow for in-depth data collection, providing rich, qualitative insights. They enable clarification of questions, probing for detailed responses, and capturing non-verbal cues. Personal interaction can build rapport, encouraging honest and comprehensive answers.

Q#50 Transformational study definition and importance

****Answer**:** A transformational study seeks to induce significant change or improvement in the subject being studied, often involving interventions aimed at altering behavior, processes, or outcomes. It is important for understanding the impact of interventions and driving meaningful progress.

Q#51 Mr. Amjad was asked in the interview about the rising price of petrol. This question leads to which condition, overstressed or other?

****Answer**:** This question could lead to an overstressed condition if Mr. Amjad is significantly affected by the rising petrol prices and feels pressure or anxiety discussing the topic. It may evoke strong emotional responses, impacting the interview's atmosphere and his ability to respond objectively.

Q#52 Define tactics used in probing

****Answer**:** Probing tactics include: 1) Silent probe, giving respondents time to think.

2) Echo probe, repeating part of the answer to encourage elaboration.

3) Neutral probe, asking non-leading follow-up questions.

4) Detailed probe, requesting specific examples or explanations.

Q#53 Write parts of the table

****Answer**:** Parts of a table include: 1) Title, which describes the table's purpose;

2) Headings, defining the columns and rows;

3) Body, containing the actual data;

4) Footnotes, providing additional information or explanations relevant to the table's content.

Q#54 Five suggestions for accurate data recording

****Answer**:** Suggestions include:

- 1) Use standardized forms or software;
- 2) Double-check entries for accuracy;
- 3) Record data immediately;
- 4) Maintain consistent coding schemes;
- 5) Regularly back up data to prevent loss, ensuring data integrity and reliability.

Q#55 Do you agree sample accessibility through email has some disadvantages?

****Answer**:** Yes, disadvantages include potential non-response bias, as not everyone checks or responds to emails. It may exclude those without internet access, affecting representativeness. Privacy concerns can also deter participation, impacting the reliability and validity of the data collected.

Q#56 Advantages of mail questionnaire

****Answer**:** Advantages include: 1) Cost-effectiveness, as it requires fewer resources than in-person surveys; 2) Convenience for respondents, who can complete it at their own pace; and 3) Wide reach, allowing researchers to collect data from geographically dispersed populations.

Q#57 Difference between control or experimental group

****Answer**:** The control group does not receive the experimental treatment, serving as a baseline. The experimental group receives the treatment. Comparing the outcomes of both groups helps determine the treatment's effect, controlling for other variables that might influence the results.

Q#58 Convenience sampling

****Answer**:** Convenience sampling involves selecting participants based on their availability and ease of access. It is quick and cost-effective but may introduce bias, as it does not represent the entire population, potentially limiting the generalizability of the findings.

Q#59 What is transformation and its importance?

****Answer**:** Data transformation involves converting data into a suitable format for analysis, such as normalizing or coding data. It is important for ensuring consistency, compatibility, and accuracy in data analysis, making it easier to interpret and draw valid conclusions from the data.

Q#60 Explain leading, loading double-barreled, Burden

****Answer**:** Leading questions suggest a particular answer, influencing responses. Loaded questions contain assumptions that may bias responses. Double-barreled questions ask about two things at once, confusing respondents. Burdensome questions demand excessive recall or detail, making them hard to answer accurately.

Q#61 Purpose of study

****Answer**:** The purpose of a study is to define its main goal, objectives, and the specific research questions it seeks to answer. It guides the research design, methodology, and analysis, ensuring the study remains focused and relevant to addressing the identified problem or hypothesis.

Q#62 Unit of analysis

****Answer**:** The unit of analysis is the primary entity being studied or analyzed in research, such as individuals, groups, organizations, or events. It determines the focus of data collection and analysis, shaping the conclusions drawn from the study.

Q#63 Study setting

****Answer**:** The study setting refers to the physical, social, or organizational context in which research is conducted. It includes locations such as laboratories, field sites, or online environments, influencing data collection methods and the generalizability of findings.

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